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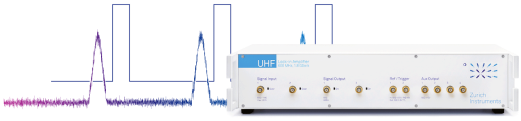
C. Clay Marston; Gabriel G. Balint-Kurti

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The Fourier grid Hamiltonian method for bound state eigenvalues and eigenfunctions

C. Clay Marston and Gabriel G. Balint-Kurti
Department of Theoretical Chemistry, The University, Bristol BS8 1TS, United Kingdom

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A new method for the calculation of bound state eigenvalues and eigenfunctions of the Schrödinger equation is presented. The Fourier grid Hamiltonian method is derived from the discrete Fourier transform algorithm. Its implementation and use is extremely simple, requiring the evaluation of the potential only at certain grid points and yielding directly the amplitude of the eigenfunctions at the same grid points.

"When one has a particular problem to work out in quantum mechanics, one can minimize the labor by using a representation in which the representatives of the most important abstract quantities occurring in the problem are as simple as possible."

P.A.M. Dirac, from *The Principles of Quantum Mechanics*, 1958.

I. INTRODUCTION

The recent work of Kosloff[1,2] has beautifully demonstrated the utility of the Fourier transform method in solving time-dependent quantum mechanical problems. The underlying reason for the power of the technique is directly related to the quotation cited above from the seminal book of Dirac. Its relevance is that the Hamiltonian operator appearing in Schrödinger's equation is composed of the sum of a kinetic energy and a potential energy operator. The kinetic energy operator is best represented in the momentum representation, as the basic vectors of this representation are eigenfunctions of both the linear momentum and the kinetic energy operators. The potential, on the other hand, is best treated in the coordinate or Schrödinger representation in which it is diagonal. Fourier transforms emerge naturally as the transformation between these two representations.

Kosloff[1,2] has utilized these ideas in evaluating the quantity $\hat{\mathscr{H}}\psi$, which is central to the time propagation process. In his work both ψ and $\hat{\mathscr{H}}\psi$ are represented as a vector whose components are the values of the function on a grid of points in coordinate space. Below we identify the matrix representation of $\hat{\mathscr{H}}$ in this vector space. We show that the calculation of the $\mathscr{H}$ matrix in this space is simple, requiring only the evaluation of the potential at the grid points and forward and reverse Fourier transforms which reduce to a summation over cosine functions. The diagonalization of the resulting Hamiltonian matrix yields the bound state eigenvalues and the eigenvectors give *directly* the amplitudes of the eigenfunctions of the Schrödinger equation on the grid points.

The Fourier grid Hamiltonian (FGH) method discussed in this paper is a special case of a discrete variable representation (DVR) method as discussed and used extensively by Light and co-workers.[3,4] Such methods were originally introduced by Harris *et al.*[5,6] They were shown to be related to Gaussian quadrature methods through a simple unitary transformation by Dickinson and Certain.[7] Another very useful technique called the discrete position operator representation, which is again closely related to the DVR methods, has also been formulated by Kanfer and Shapiro[8] and is now being widely used. The FGH method, described in detail below, has the advantage of simplicity over all the other techniques. In particular, the wavefunctions or eigenfunctions of the Hamiltonian operator are generated directly as the amplitudes of the wave function on the grid points and they are not given as a linear combination of any set of basis functions. The FGH method is variational in the same sense as most other numerical methods for calculating eigenvalues and eigenfunctions of the quantum mechanical Hamiltonian operator.[3]

Section II lays out the theoretical foundations of the FGH method. Test application to the complete set of bound state eigenvalues and eigenfunctions of a Morse curve are described in Sec. III, while Sec. IV gives a brief summary.

II. THEORY

We consider a single particle of mass m moving in one linear dimension under the influence of a potential V. The nonrelativistic Hamiltonian operator $\hat{\mathscr{H}}$ may be written as a sum of a kinetic energy and a potential energy operator

$$\hat{\mathscr{H}} = \hat{T} + V(\hat{x}) = \frac{\hat{p}^2}{2m} + V(\hat{x}) \,. \tag{1}$$

We follow here the philosophy of Dirac's book,[9] in that the operators in Eq. (1) act on vectors of an abstract Hilbert space and have not yet been cast into any particular representation. The principle representation which we will use is the Schrödinger or coordinate representation. The basic vectors or kets of this representation are denoted by $|x\rangle$ and are eigenfunctions of the position or coordinate operator $\hat{x}$:

$$\hat{x}|x\rangle = x|x\rangle \,. \tag{2}$$

The orthogonality and completeness relationships in terms of these basic vectors are

$$\langle x|x'\rangle = \delta(x - x') \tag{3}$$

and

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10 April 2024 09:03:28

$$\hat{I}_x = \int_{-\infty}^{\infty} |x\rangle\langle x| dx \,. \tag{4}$$

The potential is diagonal in the coordinate representation

$$\langle x'|V(\hat{x})|x\rangle = V(x)\delta(x-x') \,. \tag{5}$$

The eigenfunctions of the momentum operator are written as

$$\hat{p}|k\rangle = k\hbar|k\rangle \,. \tag{6}$$

The kinetic energy operator is therefore diagonal in the momentum representation

$$\langle k'|\hat{T}|\hat{k}\rangle = T_k\delta(k-k') \equiv \frac{\hbar^2 k^2}{2m}\delta(k-k') \,. \tag{7}$$

We will also require the orthogonality and completeness relationships in terms of the momentum eigenstates

$$\langle k|k'\rangle = \delta(k-k') \tag{8}$$

and

$$\hat{I}_k = \int_{-\infty}^{\infty} |k\rangle\langle k| dk \tag{9}$$

and the transformation matrix elements between the coordinate and momentum representations

$$\langle k|x\rangle = \frac{1}{\sqrt{2\pi}} e^{-ikx} \,. \tag{10}$$

Armed with these basic formulas and definitions, we may now consider the coordinate representation of the Hamiltonian operator

$$\langle x|\hat{\mathscr{H}}|x'\rangle = \langle x|\hat{T}|x'\rangle + V(x)\delta(x-x') \,. \tag{11}$$

Now inserting the identity operator (9) just to the right of the kinetic energy operator, we obtain

$$\begin{aligned}\langle x|\hat{\mathscr{H}}|x'\rangle &= \langle x|\hat{T}\left\{\int_{-\infty}^{\infty} |k\rangle\langle k|\right\}|x'\rangle dk + V(x)\delta(x-x') \\ &= \int_{-\infty}^{\infty} \langle x|k\rangle T_k\langle k|x'\rangle dk + V(x)\delta(x-x') \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{ik(x-x')}T_k dk + V(x)\delta(x-x') \,. \end{aligned} \tag{12}$$

As will be seen below, this equation is at the heart of the FGH method. The presence of the exponential factor and the integral over k may be regarded as arising from a forward, followed by an inverse Fourier transform.

A. Discretization

We wish to replace the continuous range of coordinate values x by a grid of discrete values x_i. We will use a uniform discrete grid of x values

$$x_i = i\Delta x \,, \tag{13}$$

where Δx is the uniform spacing between the grid points. It will be useful first of all to examine the discretization of the normalization integral for a wave function $\psi(x)$ [where $\psi(x)$ is the coordinate representative of a state function $\langle x|\psi\rangle = \psi(x)$]. The normalization condition for the wave function is

$$\int_{-\infty}^{\infty} \psi^*(x)\psi(x)dx = 1 \,. \tag{14}$$

Discretizing this integral on our regular grid of N values of x [see Eq. (13)], we obtain

$$\sum_{i=1}^{N} \psi^*(x_i)\psi(x_i)\Delta x = 1 \,,$$

or

$$\Delta x \sum_{i=1}^{N} |\psi_i|^2 = 1 \,, \tag{15}$$

where $\psi_i = \psi(x_i)$.

The grid size and spacing chosen in coordinate space determines the reciprocal grid size in momentum space. The total length of the coordinate space covered by the grid is $N\Delta x$. This length determines the longest wavelength and therefore the smallest frequency, which occurs in the reciprocal momentum space

$$\begin{aligned}\Delta k &= 2\pi/\lambda_{\max} \,, \\ \Delta k &= 2\pi/N\Delta x \,. \end{aligned} \tag{16}$$

This relationship gives us the grid spacing in momentum space. The central point in the momentum space grid is taken as $k = 0$, and the grid points are evenly distributed about zero. We now define an integer n by the relationship

$$2n = (N-1) \,, \tag{17}$$

where N is the (odd) number of grid points in the spatial grid. Note that the theory for an even number of grid points is just slightly more complicated.[10]

The basic bras and kets of our discretized coordinate space give the value of a wave function at the grid points

$$\langle x_i|\psi\rangle = \psi(x_i) = \psi_i \,. \tag{18}$$

The identity operator (4), which must now be compatible with Eqs. (14), (15), and (18), becomes

$$\hat{I}_x = \sum_{i=1}^{N} |x_i\rangle\Delta x\langle x_i| \,. \tag{19}$$

Similarly the orthogonality condition may now be written as

$$\Delta x\langle x_i|x_j\rangle = \delta_{ij} \,. \tag{20}$$

We now seek the discretized analog of the Hamiltonian operator matrix elements in Eq. (12).

$$\begin{aligned}H_{ij} &= \langle x_i|\hat{\mathscr{H}}|x_j\rangle \\ &= \frac{1}{2\pi}\sum_{l=-n}^{n} e^{il\Delta k(x_i-x_j)}\left\{\frac{\hbar^2}{2m}(l\Delta k)^2\right\}\Delta k + \frac{V(x_i)\delta_{ij}}{\Delta x} \\ &= \frac{1}{2\pi}\left(\frac{2\pi}{N\Delta x}\right)\sum_{l=-n}^{n} \exp[il(2\pi/N\Delta x) \\ &\quad \times(i-j)\Delta x]\{T_l\} + \frac{V(x_i)\delta_{ij}}{\Delta x} \,, \\ H_{ij} &= \frac{1}{\Delta x}\left\{\sum_{l=-n}^{n}\frac{e^{il2\pi(i-j)/N}}{N}\cdot\{T_l\} + V(x_i)\delta_{ij}\right\} \,, \end{aligned} \tag{21}$$

where

$$T_l = \frac{\hbar^2}{2m}\cdot(l\Delta k)^2 \,. \tag{22}$$

This expression for the Hamiltonian operator matrix ele-

10 April 2024 09:03:28

ments may be further simplified by combining negative and positive values of l:

$$H_{ij}=\frac{1}{\Delta x}\left\{\sum_{l=1}^{n}\frac{2\cos(l2\pi(i-j)/N)}{N}T_l+V(x_i)\delta_{ij}\right\},$$

or

$$H_{ij}=\frac{1}{\Delta x}\left\{\frac{2}{N}\sum_{l=1}^{n}\cos(l2\pi(i-j)/N)T_l+V(x_i)\delta_{ij}\right\}, \tag{23}$$

where it should be noted that $T_0=0$ [see Eq. (22)].

Armed with Eqs. (15) and (23), we can now find an expression for the expectation value of the energy corresponding to an arbitrary state function. The state function is expressed as a linear combination of the basis functions $|x_i\rangle$, which may be loosely thought of as unit Dirac delta functions distributed on the grid points

$$|\psi\rangle=\hat{I}_x|\psi\rangle=\sum_i|x_i\rangle\Delta x\psi_i\,. \tag{24}$$

The ψ_i's, which correspond to the value of the coordinate representative of the state function or the wave function at the grid points, are now the unknown coefficients to be evaluated by the variational method.[11] The expectation value of the energy corresponding to the state function $|\psi\rangle$ is

$$E=\frac{\langle\psi|\mathscr{H}|\psi\rangle}{\langle\psi|\psi\rangle}=\frac{\Sigma_{ij}\psi_i^*\Delta xH_{ij}\Delta x\psi_j}{\Delta x\Sigma_i|\psi_i|^2}. \tag{25}$$

We now define a renormalized Hamiltonian matrix

$$H_{ij}^0=\frac{2}{N}\sum_{l=1}^{n}\cos(l2\pi(i-j)/N)T_l+V(x_i)\delta_{ij}\,, \tag{26}$$

where from Eqs. (16) and (22) we see that

$$T_l=\frac{2}{m}\left(\frac{\hbar\pi l}{N\Delta x}\right)^2. \tag{27}$$

In terms of this renormalized Hamiltonian matrix, the expectation value of the energy may be written as

$$E=\frac{\langle\psi|\mathscr{H}|\psi\rangle}{\langle\psi|\psi\rangle}=\frac{\Sigma_{ij}\psi_i^*H_{ij}^0\psi_j}{\Sigma_i|\psi_i|^2}. \tag{28}$$

Minimizing this energy with respect to variation of the coefficients ψ_i yields the standard set of secular equations

$$\sum_j[H_{ij}^0-E_\lambda\delta_{ij}]\psi_j^\lambda=0\,. \tag{29}$$

The eigenvalues E_λ of this equation, which lie below the dissociation energy of the potential $V(x)$ correspond (approximately) to the bound state energies of the system. The eigenvectors ψ_j^λ give directly the (approximate) values of the normalized solutions of the Schrödinger equation evaluated at the grid points. These eigenvectors should be normalized according to Eq. (15).

B. Use of the fast Fourier transform technique

A state function $|\psi\rangle$ may be represented as a vector on a discretized grid of points either in coordinate space, as in Eq. (24), or in momentum space. These two alternative representations of the same state function may be written as

$$|\psi\rangle=\sum_i|x_i\rangle\Delta x\psi(x_i)=\sum_i|x_i\rangle\Delta x\psi_i^x \tag{30}$$

or

$$|\psi\rangle=\sum_l|k_l\rangle\Delta k\psi(k_l)=\sum_l|k_l\rangle\Delta k\psi_l^k\,. \tag{31}$$

The fast Fourier transform technique (FFT)[10,13–15] provides a uniquely efficient method of transforming from one of these representations to the other. It is not our purpose here to describe the FFT technique itself, but only to outline how it may be used in the present context.

The FFT technique may be represented as a unitary matrix transformation between the coordinate and momentum grid representations of the state function. Denoting the matrix which represents the forward FFT by $\mathscr{F}$, we may write

$$\psi^k=\mathscr{F}\psi^x\,, \tag{32}$$

where both ψ^k and ψ^x are column vectors.

Let us now define a column vector ϕ_n which is composed entirely of zeros except for a single element of unity in the nth row

$$\phi_n=\begin{pmatrix}0\\0\\\vdots\\1\\\vdots\\0\\0\end{pmatrix}\leftarrow n\text{th row}\,. \tag{33}$$

We may now use this vector, together with a forward and reverse FFT transformation, to give the nth column of the Hamiltonian matrix H^0 [see Eq. (26)]

$$H_{in}^0=[\mathscr{F}^{-1}T\mathscr{F}+V)\phi_n]_i\,, \tag{34}$$

where T and V are the diagonal kinetic energy [Eq. (27)] and potential energy [$V(x_i)$] matrices. By repeating this process for all the possible N vectors ϕ_n, we can generate the complete matrix H^0.

The form of the Hamiltonian matrix elements given in Eq. (34) is the same as that which has been used in discussions of the DVR and related methods[3,7,8] and shows clearly the relationship of the present FGH method to other discrete variable representation techniques.

III. NUMERICAL IMPLEMENTATION

Although the stationary states of the Morse potential admit to analytic solutions,[11,12] difficulties are often reported in the calculation of the higher levels. We have chosen to compare the bound state eigenvalues and eigenfunctions, computed by diagonalizing the Hamiltonian constructed following the FGH formulation given in Eq. (26), with eigenvalues and eigenfunctions calculated from the analytic solution to the Morse curve problem.[12] The equation of a Morse potential is

$$V(x)=D[1-e^{-\beta(x-x_e)}]^2. \tag{35}$$

The parameters used are those applicable to H_2 and are given in Table I. The reduced mass of H_2 was also used in the construction of the Hamiltonian operator matrix elements.

10 April 2024 09:03:28

TABLE I. Parameters of the Morse curve.

$D = 0.1744$ a.u. $= 4.7457$ eV
$\beta = 1.02764$ a.u. $= 1.94196 \times 10^{10}$ m^{-1}
$x_e = 1.40201$ a.u. $= 0.74191 \times 10^{-10}$ m

In order to compute all bound states, the potential must be sampled with a sufficient density of points and over a suitable magnitude of displacement of the interatomic spacing. The parameter $\gamma = \sqrt{2D\mu}/\beta = 17.41$ provides an estimate of the number of bound states contained in the potential. Based on this estimate, the grid points were uniformly distributed between $x = 0$ and 1.5 times the outer classical turning point of the 16th eigenstate.

As stated in the previous section, the number of sampling points in both coordinate and momentum space must be odd to insure that $T_0 = 0$ is the central point of a spectrum consisting of paired values of equal index magnitude, but of opposite sign. The grid pints in coordinate space are given by

$$x_i = i\Delta x, \quad 0 \leqslant i \leqslant N-1, \tag{36}$$

where $\Delta x = L/N$ and L is the length of the range of x values sampled. While those in momentum space are given by

$$k_l = l\Delta k, \quad -n \leqslant i \leqslant n, \tag{37}$$

where $\Delta k = 2\pi/L$.

The Hamiltonian was then constructed for values of N of 65 and of 129 as in Eq. (26) and diagonalized with the EISPACK subroutine RS using double precision on an IBM 3090 computer. The results are presented in Table II. The exact analytic eigenvalues are given in the final column and eigenvalues obtained from diagonalization of Hamiltonian matrices of order 65 and 129, respectively, appear in the preceding two columns. For both 65 and 129 grid points, the lowest eigenvalue differs by only 10^{-8} a.u. (3×10^{-7} eV) from that of the exact analytic values. For the highest eigenvalue, the differences are, respectively, 3×10^{-5} and 2×10^{-7} a.u. for the two differently sized grids. As may be seen from the table, the eigenvalues converge from above on their exact analytic values.

TABLE II. Comparison of eigenvalues calculated using the Fourier grid Hamiltonian method and exact analytic formula for a Morse potential representing H_2. The full set of bound state eigenvalues are listed.

Vibrational quantum number v	Eigenvalues calculated using the Fourier grid Hamiltonian method (a.u.) $N = 65$	$N = 129$	Exact analytic eigenvalues (a.u.)
0	0.009 869 23	0.009 869 23	0.009 869 22
1	0.028 745 36	0.028 745 36	0.028 745 35
2	0.046 471 71	0.046 471 73	0.046 471 72
3	0.063 048 33	0.063 048 35	0.063 048 33
4	0.078 475 64	0.078 475 20	0.078 475 18
5	0.092 753 67	0.092 752 29	0.092 752 27
6	0.105 879 09	0.105 879 62	0.105 879 60
7	0.117 846 86	0.117 857 19	0.117 857 17
8	0.128 661 57	0.128 685 00	0.128 684 98
9	0.138 340 83	0.138 363 06	0.138 363 03
10	0.146 896 50	0.146 891 35	0.146 891 32
11	0.154 317 41	0.154 269 87	0.154 269 85
12	0.160 579 03	0.160 498 64	0.160 498 62
13	0.165 665 18	0.165 577 65	0.165 577 63
14	0.169 577 79	0.169 506 90	0.169 506 89
15	0.172 330 36	0.172 286 39	0.172 286 38
16	0.173 944 40	0.173 916 37	0.173 916 11

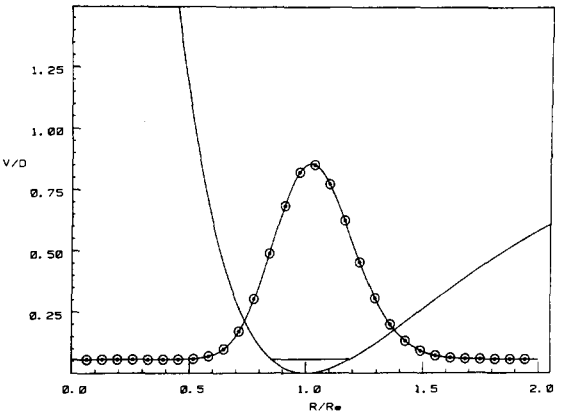


FIG. 1. Comparison of analytic (solid line) and numerically computed (dots surrounded by circles) FGH method eigenfunctions for the ground vibrational state of the Morse potential. Parameters and masses used correspond to the H_2 molecule. The wave functions are superimposed on the Morse potential. See the text for further details.

Figures 1–3 compare the computed and analytic eigenfunctions obtained from the 129 point grid for eigenstates corresponding to vibrational quantum numbers $v = 0$, 5, and 15. The circles with dots in their centers represent the wave functions calculated by the FGH method, while the solid lines are obtained from the analytic solutions.[12] The wave functions are superimposed on the scaled Morse potential curve with the zero of the wave functions placed at the bound state energies. The analytic solutions have been scaled to coincide with the numerically computed ones at their

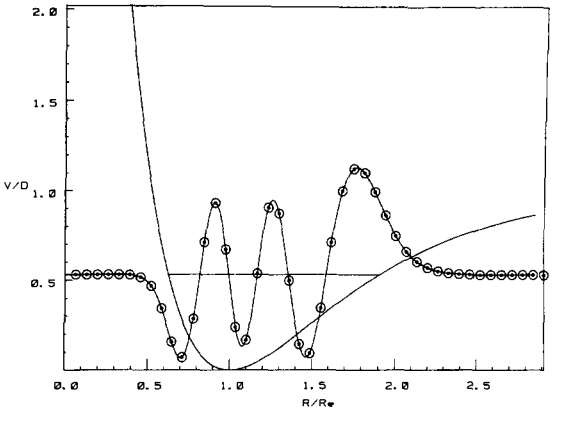


FIG. 2. Comparison of analytic (solid line) and numerically computed (dots surrounded by circles) FGH method eigenfunctions for the $v = 5$ vibrational state of the H_2 Morse potential. Details are as in Fig. 1 and the text.

10 April 2024 09:03:28

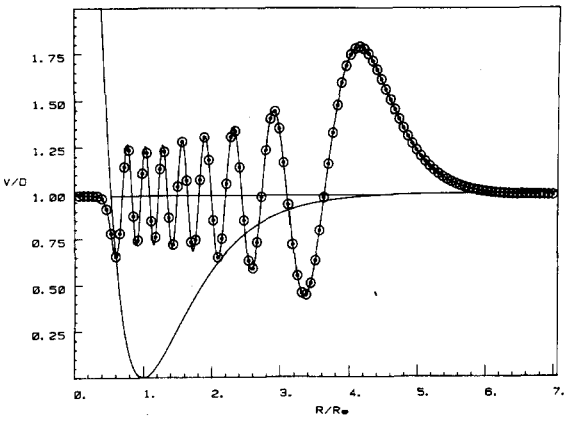


FIG. 3. Comparison of analytic (solid line) and numerically computed (dots surrounded by circles) FGH method eigenfunctions for the $v = 15$ vibrational state of the H_2 Morse potential. Details are as in Fig. 1 and the text.

maximum values. This was done for expediency so as to avoid the need to compute the rather awkward analytic normalization constants. Also the normalization condition $\Sigma_i |\psi_i|^2 = 1$ has been used, rather than that of Eq. (15). The differences between the FGH and analytic wave functions are nowhere discernible to the resolution of the figures.

IV. SUMMARY

The Fourier grid Hamiltonian method, formulated and tested in this paper, is an extremely simple numerical variational technique for calculating the bound state eigenvalues and eigenfunctions of the Schrödinger equation. The key equations of the method are Eqs. (26) and (27), which give explicitly the form of the Hamiltonian matrix. The eigenvectors of this matrix yield directly the values of the wave functions evaluated at the grid points.

Our calculations have clearly demonstrated that the method can yield highly accurate eigenvalues and eigenvectors. As with any numerical method, there are a number of parameters which can be varied to enhance the accuracy of the results. There are, however, very few of these in our present FGH method, consisting in only the length and position of the section of coordinate space sampled and the number of grid points used. Its transparency and utilitarian simplicity is clearly one of the strong points of the method.

The FGH method arises out of the work of Kosloff and co-workers[1,2] on time-dependent methods in quantum mechanics. Kosloff and Tal-Ezer[2] have studied a closely related time-dependent relaxation method, which generates the eigenfunctions one at a time and relies on projecting out the lower-lying eigenfunctions to obtain higher ones. The present method seems more straightforward when several eigenvalues and eigenfunctions are required.

In common with other grid based[2] or discrete variable representation[4,7,8] methods, no matrix elements of the potential between a set of basis functions are required. The potential must simply be evaluated at certain values of the coordinates, in this case the grid points. The method is simpler than most DVR type methods in that no explicit matrix transformations are required and no model potential, with its own set of parameters and basis functions, is used.

A shortcoming of the method, especially when its generalization to many mathematical dimensions is considered, is the high order of the matrices which must be diagonalized. The FGH method involves the use of matrices of order $(N \times N)$, where N is the number of grid points used. In one dimension, this does not constitute a problem for modern computers. The method, and possible variations of it using modern iterative procedures for finding eigenvalues and eigenvectors,[16–18] is well suited for the use with vector and parallel processing techniques. Developments in this direction, when coupled with the great simplicity of the basic formalism, may well render the FGH method useful also for multidimensional problems.

The method generates wave functions in a very convenient form for subsequent use as basis functions over which a numerical quadrature is to be performed. In their recent work on the calculation of bound state eigenfunctions for systems possessing large amplitude molecular vibrations Bačič *et al.*[4] have advocated using a set of distributed Gaussian basis (DGB) functions in one or two dimensions coupled with the DVR method in the other dimensions. The DGB functions are used to solve a one-dimensional Schrödinger equation and the lowest few resulting eigenfunctions are used as a truncated basis for the degree of freedom involved. Our present FGH method also provides a convenient way to generate such one-dimensional bases for use in multidimensional problems in conjunction with DVR or other[19,20] methods.

After submission of our paper, two methods related to the FGH method described herein have been brought to our attention. Guardiola and Ros[21] have discussed a grid method based on the use of a sine basis, and Light and coworkers[22,23] have discussed the use of the DVR method using a Chebychev basis. As the arguments of the Chebychev polynomials are cosines of the internuclear separation,[22] this latter method may be equivalent to the use of a cosine basis. Our FGH method and both of these other two methods all use a regularly spaced set of grid points. They all share the property that in the grid representation the components of the eigenvectors of the Hamiltonian matrix are proportional to the values of the wave functions at the grid points.

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10 April 2024 09:03:28